



A.M.($\leq$)	Any	Moon P.	Any/Bright
Image Quality		Any seeing os	
Cloudy Cover		Any	

Title of Proposal: Astrometry of Jupiters Galilean Satellites

Telescope 0.60m Boller & Chivens

Abstract:

Astrometry of the Galilean satellites deepens our understanding of orbital dynamics and gravitational interactions in the Jovian system, contributing to planetary geology and the history of the Solar System. Precise measurements of the positions of Io, Europa, Ganymede, and Callisto improve orbital models and enable the investigation of tidal effects (important sources of internal heating, especially in Europa and Io) which influence geological processes and the possible existence of subsurface oceans. Continuous astrometric monitoring is essential to maintain accurate ephemerides and reduce uncertainties in orbital predictions. These data are also crucial for the interpretation of missions such as JUICE and Europa Clipper, which will provide new insights in the coming years. Together, these results can reveal aspects of the geological evolution of these bodies, directly linked to tidal forces, and significantly advance planetary exploration.

Category: Solar System: Small Bodies (Asteroids, Comets, Moons, Kuiper Belt, Near Earth Objects)

Time requested: 36 hours.

Minimum time accepted: 36 hours.

Observing Mode: Classical Remote (CR)

Program Type: Regular Proposal (RP)

Principal Investigator (PI): Viviane Figueiredo Peixoto

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Is there a thesis or dissertation that will benefit from the data?: YES

Student	E-mail	Degree	Defense
Viviane Figueiredo Peixoto	vivianepeixoto989@gmail.com	PhD student	02/2028

Optimal dates:

November 02, 03, 04

Impossible dates:

Other

Instruments:

Imager (CAM 1; 2 or 4)

Detectors - Ikon 2048 x 2048

Scientific Justification

The Galilean satellites—Io, Europa, Ganymede, and Callisto—are key objects for understanding the interplay between orbital dynamics and planetary geology in the Solar System. Their orbital evolution is strongly influenced by mutual gravitational interactions and tidal forces exerted by Jupiter, which play a central role in driving internal heating, geological activity, and, in the case of Europa, the maintenance of a subsurface ocean. Despite decades of study, important uncertainties remain in their orbital parameters and in the quantitative characterization of tidal dissipation within the Jovian system. This project aims to obtain high-precision astrometric measurements of the Galilean satellites in order to refine their orbital models and investigate the role of tidal forces in shaping their long-term evolution.

The proposed observations will employ an alternative astrometric technique based on the combination of consecutive images. Due to the strong brightness of Jupiter and its satellites, classical astrometry is limited by the scarcity of reference stars in individual frames, typically yielding positional accuracies on the order of 100 mas[1][2]. Our method overcomes this limitation by using a bright reference star (magnitude ≤ 10) present in the field to align a sequence of short-exposure images. The co-addition of these aligned frames increases the signal-to-noise ratio and enables the detection of fainter reference stars (down to magnitude ~ 14), allowing for a significant improvement in astrometric precision, reaching ~ 20 mas[3]. The reduced data will be processed using standard calibration procedures and astrometric reduction pipelines, followed by dynamical analysis to assess the contributions of tidal interactions to the observed orbital behavior.

The scientific impact of this project is twofold. First, it will provide new high-precision astrometric data that contribute to the continuous improvement of ephemerides for the Galilean satellites[4], which are essential for both ground-based observations and space mission planning. Second, by enabling more accurate dynamical modeling, these results will help constrain the magnitude of tidal dissipation in Jupiter and its satellites, with direct implications for their internal structures and thermal histories. This is particularly relevant in the context of upcoming missions such as JUICE and Europa Clipper, which will rely on accurate orbital knowledge for navigation and scientific interpretation. More broadly, this project contributes to advancing our understanding of tidal processes and their role in planetary system evolution[5].

Experimental Design

The target selection is constrained by the requirement that the Jovian system shares the field of view with at least one reference star brighter than magnitude 10, which is essential for the image-combination astrometric technique. The selected observing dates (November 2nd, 3rd, and 4th, 2026) correspond to optimal configurations in which such stars are present within the $12' \times 12'$ field provided by the Ikon CCD on the Boller & Chivens telescope. This strategy maximizes the number of detectable reference stars after image combination and allows, whenever possible, the simultaneous observation of multiple Galilean satellites within the same field.

We request a total of three nights, with observations concentrated during the second half of each night, when the Jovian system is above a suitable altitude (starting at ~04:00 local time). Individual exposures will range from 4 to 20 seconds, depending on observing conditions, particularly seeing and transparency. For each observing sequence, hundreds of consecutive images will be acquired to enable robust frame alignment and stacking. The methane filter will be used to suppress the scattered light from Jupiter, reducing background contamination and preventing saturation while preserving adequate signal from the satellites and reference stars.

The primary observables are the precise astrometric positions of the Galilean satellites. These measurements directly address the scientific goals by enabling orbital refinement and dynamical studies, particularly those related to tidal interactions. Calibration frames (bias and flat-fields) will be obtained and applied using standard procedures within IRAF/PyRAF environments. Image combination will be performed using a dedicated Python-based pipeline, and astrometric reduction will be carried out with the PRAIA package[6]. A critical step in the process is the accurate alignment of consecutive frames using the bright reference star, which ensures the improvement in signal-to-noise ratio and the detection of fainter astrometric references[7].

The observations are designed to achieve a signal-to-noise ratio (S/N) greater than 800 for the satellites and greater than 20 for the reference star in individual frames, ensuring centroiding precision at the level of ~0.05 pixels, corresponding to ~20 mas. After image stacking, fainter stars down to magnitude ~14 become measurable, significantly improving the astrometric solution. The observations are not strongly constrained by lunar phase, as the targets are bright, and all moon phases are acceptable. Typical seeing conditions at OPD (FWHM ~1–2") are sufficient for the required precision, and observations will be conducted at low airmass (< 2) to minimize atmospheric effects. No strict cloud-free (photometric) conditions are required, but stable transparency is preferred to ensure consistent image quality across sequences.

This observational setup is fully aligned with the scientific objectives, ensuring the acquisition of high-precision astrometric data suitable for advancing dynamical models of the Galilean satellite system.

Technical Description

The proposed observations will be conducted using the Boller & Chivens 0.6 m telescope equipped with the Ikon CCD camera, which provides a field of view of approximately $12' \times 12'$. This configuration is essential for the success of the project, as it maximizes the probability of including both the Galilean satellites and at least one bright reference star (magnitude ≤ 10) within the same field (Figure 1). The use of the methane filter is critical to suppress the scattered light from Jupiter, significantly reducing background contamination and preventing detector saturation, while preserving sufficient signal from the satellites and reference stars for high-precision astrometry.

The observational constraints are relatively flexible due to the brightness of the targets. All lunar phases are acceptable, as the Galilean satellites (typical magnitudes ~ 5) are significantly brighter than the sky background even under full Moon conditions. The observations will be carried out at airmass < 2 to minimize atmospheric refraction and distortion effects, ensuring accurate centroid determination. Typical seeing conditions at the Observatório do Pico dos Dias (FWHM $\sim 1\text{--}2''$) are sufficient to achieve the desired astrometric precision. While fully photometric conditions are not required, stable transparency throughout each observing sequence is important to ensure consistent image quality for the stacking process.

Exposure times will range from 4 to 20 seconds, depending on atmospheric conditions and the instantaneous brightness of the targets. These values are based on previous observing runs and are optimized to avoid saturation of the satellites while maintaining adequate signal for the reference star. Individual frames are expected to reach a signal-to-noise ratio (S/N) greater than 800 for the satellites and greater than 20 for the reference star. This ensures centroiding precision at the level of ~ 0.05 pixels (~ 20 mas). The combination of hundreds of consecutive frames further increases the effective S/N, enabling the detection of fainter reference stars down to magnitude ~ 14 , which is essential for robust astrometric solutions.

We request a total of three nights, with observations concentrated in the second half of each night ($\sim 4\text{--}6$ hours per night), when the Jovian system is observable at suitable altitude. Each night will include a short initial setup period (~ 30 minutes) for telescope configuration and focusing, followed by continuous science observations. Calibration frames (bias and flat-fields) will be obtained at the end of the night (~ 30 minutes). The remaining time will be efficiently used for science exposures, with negligible overhead between consecutive frames due to the short exposure times and fast CCD readout. This observing strategy ensures that the requested time is sufficient to acquire the necessary data volume for high-precision astrometric analysis.

Overall, the combination of instrument configuration, observing strategy, and exposure time estimates demonstrates that the proposed program is technically feasible and well-matched to the scientific goals.

Figures and Tables

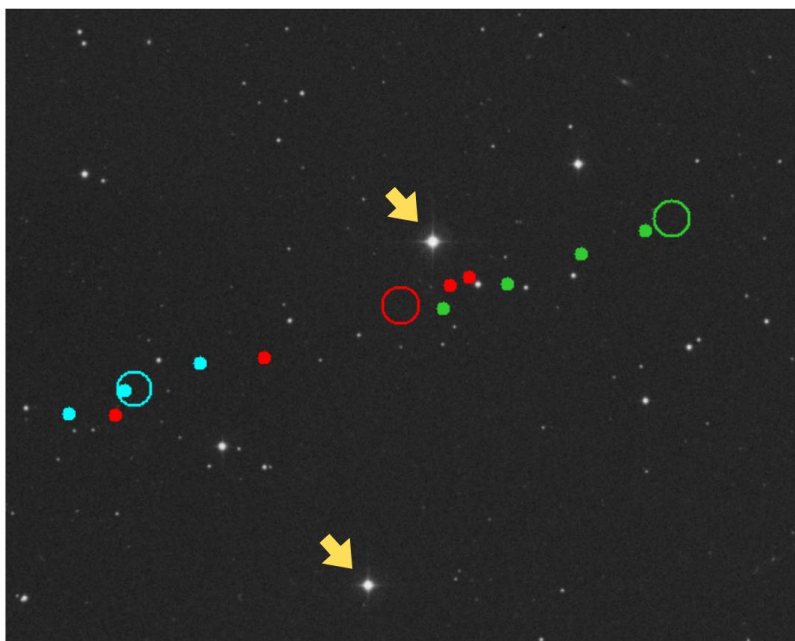


Figure 1: Position of Jupiter and the Galilean satellites on November 2nd, 3rd, and 4th, 2026 (shown in green, red, and blue, respectively). The yellow arrow indicates the stars that will be used for image combination.

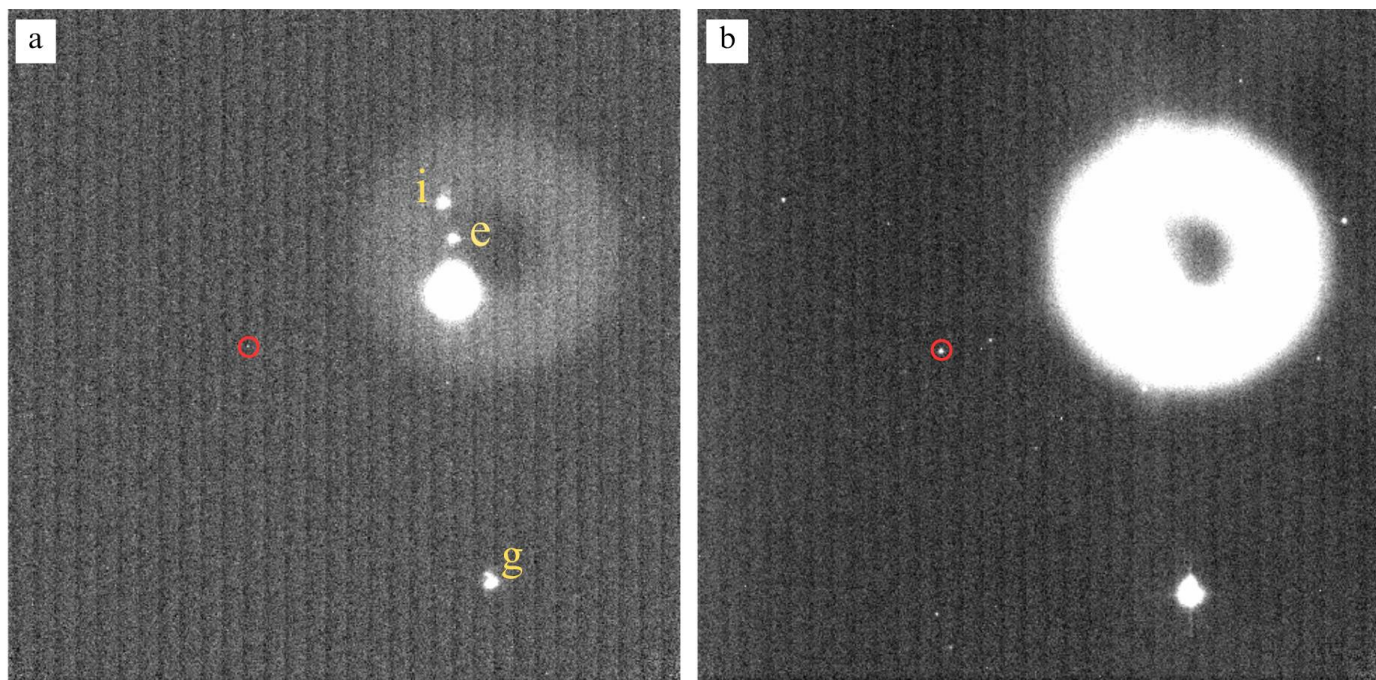


Figure 2: Example of the application of the image-combination technique. (a) Image obtained on the night of March 2nd, 2025, with the Boller & Chivens telescope, using a 20-second exposure with the methane filter. The red circle highlights the star used as the reference for image combination. The satellites visible in the field are indicated by their initials (i – Io; e – Europa; g – Ganymede). (b) Resulting image from the combination of 75 images obtained during that night, allowing a significant increase in the number of reference stars in the field.

References

- [1] Kiseleva, T. P., Devyatkin, A. V., Khovritchev, M. Yu., et al. (2008). The astrometric observations of planetary satellites, close approaches and occultations of stars by asteroids and mutual events in the systems of planetary satellites with the 26-in. refractor of Pulkovo Observatory in 1995–2006. *Planetary and Space Science*, 56(14), 1908–1912.
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- [6] Assafin, M. (2023). Astrometry with PRAIA. *Planetary and Space Science*, 238, 105801.
- [7] Lainey, V., Jacobson, R. A., Tajeddine, R., et al. (2017). New constraints on Saturn's interior from Cassini astrometric data. *Icarus*, 281, 286–296.
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- [9] Morgado, B., Assafin, M., Vieira Martins, R., et al. (2019). APPROX – mutual approximations between the Galilean moons: the 2016–2018 observational campaign. *Monthly Notices of the Royal Astronomical Society*, 482, 5190–5200.
- [10] Morgado, B., Assafin, M., Vieira Martins, R., et al. (2019). First stellar occultation by the Galilean moon Europa and upcoming events between 2019 and 2021. *Astronomy & Astrophysics*, 626, L4.
- [11] Morgado, B., Assafin, M., Vieira Martins, R., et al. (2019). The 2014–2015 Brazilian mutual phenomena campaign for the Jovian satellites and improved results for the 2009 events. *Planetary and Space Science*, 179, 104736.
- [12] Morgado, B. E., Gomes-Júnior, A. R., Braga-Ribas, F., Vieira-Martins, R., Desmars, J., Lainey, V., ... & Winter, O. C. (2022). Milliarcsecond astrometry for the Galilean moons using stellar occultations. *The Astronomical Journal*, 163(5), 240.

Previous missions of this proposal to this telescope

This proposal is part of an ongoing PhD research project conducted at the Observatório do Valongo (UFRJ), focused on the astrometry of the Galilean satellites. Previous observing time for this project was allocated on the Boller & Chivens telescope at the Observatório do Pico dos Dias (OPD) during the 2025A and 2025B semesters.

In semester 2025A, a total of three nights were granted. Of these, two nights were successfully executed, while one was lost due to unfavorable weather conditions. During the successful nights, it was possible to acquire a substantial number of images of the Galilean satellites under suitable observing conditions. These data were used to test the image-combination technique proposed in this project. Preliminary results demonstrate that the method is effective in improving the signal-to-noise ratio and increasing the number of detectable reference stars in the field. The resulting astrometric precision is consistent with expectations, reaching values close to ~ 20 mas. Data reduction for these nights has been partially completed, including calibration (bias and flat-field correction) and initial astrometric processing.

In semester 2025B, six nights were allocated; however, all scheduled observations were lost due to poor weather conditions. As a result, no additional data were obtained during this period.

The data acquired in 2025A are of sufficient quality to support the feasibility of the proposed methodology (Figure 2). No significant instrumental issues were encountered, and the observing strategy proved to be well-suited to the scientific goals. The main limitation identified was the dependence on specific sky configurations, requiring the presence of a sufficiently bright reference star within the field of view. This constraint has been carefully considered in the planning of the current proposal, with observing dates selected to maximize the likelihood of meeting this requirement.

Overall, the previous observing runs validate both the technical approach and the scientific potential of the project, justifying the request for additional observing time to expand the dataset and improve the robustness of the astrometric and dynamical analyses.

Previous missions of other proposals to this telescope

The Boller & Chivens telescope at the Observatório do Pico dos Dias (OPD) has been extensively used by our research group in several independent but related proposals focused on astrometry of Solar System bodies, including the Galilean satellites and small bodies such as asteroids and trans-Neptunian objects. These programs have been conducted over multiple semesters and have consistently demonstrated the suitability of this facility for high-precision astrometric measurements.

In particular, previous proposals led by members of this collaboration (e.g., semesters between 2016 and 2024) have successfully employed similar observational strategies, including the use of short-exposure imaging and astrometric reduction with the PRAIA package. These observations resulted in multiple peer-reviewed publications (e.g. references [8-12]), including studies of mutual approximations between Galilean satellites, stellar occultations, and campaigns of mutual phenomena. The data obtained in these programs have achieved astrometric precisions at the level of a few tens of milliarcseconds, confirming the capability of the telescope and instrumentation to meet the scientific requirements of this proposal.

Data reduction in these previous programs is fully mature, relying on well-established pipelines that include standard calibration (bias and flat-field correction), astrometric solution using reference star catalogs, and

quality control procedures. The resulting datasets have been successfully used to refine orbital parameters, improve ephemerides, and support predictions of stellar occultations. These outcomes demonstrate that the data quality obtained with this telescope is fully adequate to achieve the intended scientific goals.

Some limitations have been identified in past observations, primarily related to weather losses and the difficulty of obtaining sufficient numbers of reference stars in individual frames when observing bright targets such as the Galilean satellites. These challenges motivated the development of alternative techniques, such as the image-combination method proposed in this project, which directly addresses these issues by increasing the effective number of reference stars and improving the signal-to-noise ratio.

Overall, the extensive experience accumulated by the team using the Boller & Chivens telescope, along with the successful scientific results obtained in previous proposals, strongly supports the feasibility and expected impact of the present project.

Previous results in field by the Principal Investigator (PI)

The Principal Investigator, Viviane Figueiredo Peixoto, is currently a PhD candidate in Astronomy at the Observatório do Valongo (UFRJ), working at the interface between planetary geology and orbital dynamics, with particular emphasis on tidal processes in icy satellites. Her research focuses on connecting surface geology with the dynamical evolution of planetary systems through quantitative approaches.

As part of her doctoral work, the PI has conducted a topological analysis of fracture networks on Europa, applying statistical and geometric methods to characterize fracture patterns and infer the stress regimes shaping the icy shell. This work resulted in a peer-reviewed publication in *Icarus*, contributing to the understanding of tectonic processes and tidal deformation in icy moons. She is also currently preparing a second article, based on research developed during her stay at the Jet Propulsion Laboratory (JPL), extending these methods to other icy bodies to investigate analogies between different satellite systems.

A key strength of this proposal lies in the extensive experience of the PI's advisor, Prof. Dr. Bruno Morgado, whose work has significantly advanced the field of high-precision astrometry of Solar System bodies, particularly the Galilean satellites [3]. His research has established robust methodologies for extracting precise astrometric measurements from ground-based observations, including the use of mutual approximations, stellar occultations, and coordinated observational campaigns [8-12]. These efforts have led to substantial improvements in orbital solutions and ephemerides, demonstrating the capability of ground-based techniques to achieve accuracies at the level of tens of milliarcseconds. The PI is directly involved in this research line and benefits from this well-established framework in astrometry and celestial mechanics.

Although the PI's primary expertise lies in planetary geology, this proposal represents a natural extension of her research toward observational constraints, particularly through high-precision astrometry of the Galilean satellites. The proposed image-combination technique builds upon and complements the methodologies developed by Morgado and collaborators, aiming to overcome limitations related to the scarcity of reference stars and further improve astrometric precision.

Overall, the combination of the PI's background in planetary geology and the advisor's extensive experience in astrometry provides a strong and complementary foundation for the proposed project, ensuring both its scientific relevance and technical robustness.

PI and Co-Is publications relevant to this proposal

- Assafin, M. (2023). *Astrometry with PRAIA*. Planetary and Space Science, 238, 105801.
- Catani, L. M., Assafin, M., Morgado, B. E., Santos-Filho, S., Braga-Ribas, F., Vieira-Martins, R., ... & Botelho, R. B. (2023). The 2021 mutual phenomena involving the Galilean satellites of Jupiter and the inner satellite Thebe. *Monthly Notices of the Royal Astronomical Society*, 526(4), 6145–6154.
- Lainey, V., Arlot, J.-E., Karatekin, Ö., & Van Hoolst, T. (2009). Strong tidal dissipation in Io and Jupiter from astrometric observations. *Nature*, 459, 957–959.
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- Peixoto, V. F., et al. (2026). *Topological analysis of Europa's fracture networks*. Icarus.

Objects List.

Object	A.R.(h,m)	Dec.(deg)	Mag.	Band	Type	Comments
Jupiter_02NOV2026	09:46:35.85	+14:08:28.2	0	V	Sci	Mag -2
Jupiter_03NOV2026	09:47:03.49	+14:06:19.6	0	V	Sci	Mag -2
Jupiter_04NOV2026	09:47:30.56	+14:04:13.8	0	V	Sci	Mag -2